

GNN Project

AAAI Press

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Next Steps Toward an AAAI Submission

The current report establishes a clean baseline and an honest evaluation protocol, but the gap between what is shown here and an AAAI-caliber submission is non-trivial. This section lays out the architectural redesign, baseline matrix, experimental plan, and timeline required to close that gap.

Reframing the Contribution

The framing “GNN adapter beats MLP adapter on CheXpert” is too narrow, and the empirical margin in Section 6 is too small to defend against a competitive reviewer pool. We re-cast the contribution along three pillars:

1. A **structured adapter architecture** that genuinely exploits inductive structure. The present bipartite head, with seven attribute nodes and one pooled object node at depth two, is mathematically close to an MLP over $[e_{CLIP}; z; p]$, and a reviewer will demonstrate this in the rebuttal.
2. A **cross-site generalization** result showing that the adapter is more portable across hospitals than parameter-efficient fine-tuning of the VLM itself — the practical claim that matters to clinical deployment.
3. The **leakage-free calibration protocol** formalized as a reusable contribution, accompanied by a small consistency proposition (§).

This framing survives the four obvious reviewer questions: *Why not LoRA?* (we match or beat it), *Why structure?* (the structured attention performs work an MLP cannot), *Does it generalize?* (cross-site experiments), and *Is it just calibration?* (a calibration-only baseline is reported separately).

Architecture

We propose three candidates: the first is the primary architecture, the second a layered prior, the third a strong secondary experiment.

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Option A — Concept-Evidence Cross-Attention Adapter (primary). Drop the single pooled CLIP embedding. Instead consume the full sequence of frozen image tokens from a vision encoder (CLIP-ViT, BiomedCLIP, RAD-DINO, or an MAE-CXR variant) — approximately 196 spatial tokens for ViT-B/16 at 224^2 . Treat the C clinical concepts as learnable query embeddings initialized from the text-encoder embeddings of the label names so they carry semantic structure from initialization. A small cross-attention block (one to two heads, two layers) lets concept queries attend over the image tokens; the attended evidence is combined with VLM logits through a learned gating function. This architecture is genuinely structured: it produces per-finding spatial attention maps (a trust-and-explainability feature radiologists weight heavily), it cannot be reproduced by an MLP over a pooled embedding, and the parameter count remains under one million. Implementation reuses `nn.MultiheadAttention` and Hugging Face cross-attention modules.

Option B — Heterogeneous concept graph with Rad-Graph priors. Replace the empirical co-error adjacency with the radiology knowledge graph extracted from MIMIC-CXR reports (?). Concept nodes span the fourteen CheXpert findings, latent parent concepts (*Lung Opacity*, *Cardiac Silhouette*), and modifier nodes (location, severity). This is principled, defensible to a reviewer, and lets the graph carry actual medical structure rather than VLM-error statistics.

Option C — Multi-VLM ensemble adapter. Nodes are $(VLM \times Label)$ tuples; the adapter learns per-class which frozen scorer to trust. This is a practical production scenario (multi-provider APIs) and is exactly the setting in which bipartite structure can pay off.

Recommendation. Build Option A as the main architecture, layer Option B’s RadGraph prior as a separately ablatable component, and present Option C as the applied-systems experiment. Drop “GNN” from the working title; *Concept-Evidence Adapters for Frozen Vision-Language Models on Multi-Label Medical Classification* is more precise. A seven-node “GNN” invites a taxonomy attack we should not concede in round one.

Baseline Matrix

Every reviewer in this space mentally compiles the following list. The submission must report every row.

Frozen-output baselines. Zero-shot VLM (already in place); linear probe on $[z; p]$; MLP on the same; MLP with pooled CLIP appended; a faithful re-implementation of ML-GCN (?) on our label graph; the existing bipartite head, retained as the prior version of the work.

Parameter-efficient fine-tuning baselines. LoRA on Qwen2-VL at ranks $\{4, 8, 16\}$; LoRA restricted to the vision encoder; prefix tuning; visual prompt tuning. This is the load-bearing comparison: if LoRA-16 outperforms our adapter, the paper changes shape.

Medical-domain encoder baselines. Linear probes on BiomedCLIP, RAD-DINO, and an MAE-CXR checkpoint where available. “Frozen CLIP-B/32 + adapter” is not a fair fight; “frozen BiomedCLIP + adapter” is.

Loss and calibration baselines. Asymmetric loss (?) on the same backbone; logit adjustment (?); temperature scaling; per-class isotonic regression via `sklearn.isotonic`. These quantify how much of any reported gain is loss-function engineering versus architecture.

Experimental Matrix

Datasets. CheXpert alone is insufficient. We add MIMIC-CXR-JPG (PhysioNet credentialed access) and NIH ChestX-ray14 at minimum, and PadChest if time allows. Train on CheXpert; evaluate in-distribution on CheXpert test; then cross-site on MIMIC-CXR and NIH. The cross-site numbers are the single most important result; they justify the portability claim.

Metrics. Per-class AUROC and AUPRC, F1 at the calibrated operating point, and calibration diagnostics (ECE, Brier score) per class. Macro-averaged in the headline table; per-class in an appendix table. All numbers reported as mean $\pm$ standard deviation over five seeds together with a paired bootstrap 95% confidence interval on the test predictions (1,000 resamples is sufficient at $\sim 9,000$ rows). For AUROC comparisons specifically, run DeLong’s test for paired ROC differences. Reuse `torchmetrics.classification.MultilabelAUROC` and `sklearn.metrics.roc_auc_score`, cross-checked against CheXpert reference numbers in a dedicated `tests/` module.

Inference cost as a first-class metric. Report GPU-hours to train each baseline, total parameter count, inference latency, and peak GPU memory. This is the central practical claim of the paper; without it the “cheap adapter” story has nothing to stand on. Present a cost table adjacent to the accuracy table in Section 1.

Ablations that earn their space. Image encoder swap (CLIP-B/32 $\rightarrow$ BiomedCLIP $\rightarrow$ RAD-DINO); concept-query initialization (random vs. label-text encoding); graph prior (none, co-occurrence, co-error, RadGraph); number of cross-attention layers; calibration method (none, threshold, temperature, isotonic). Two further controls deserve placement in the main paper rather than the appendix: a **permuted-graph control** (if the model is unchanged under random edge shuffling, the graph is doing nothing) and a **held-out-concept test** measuring zero-shot generalization to labels masked during training.

Principled Angle

A small theoretical statement raises the ceiling on a primarily empirical AAAI submission. Two inexpensive options:

- **Sample-complexity statement.** Bound the number of labeled examples required for the adapter to recover the Bayes-optimal classifier given frozen scores of a stated quality. This follows from the linearity of the residual head and a standard linear-probe argument; it pins down the regime in which our adapter dominates fine-tuning.
- **Calibration-consistency statement.** Show that the leakage-free four-way protocol yields unbiased per-class threshold estimates whose error decays at the expected $O(1/\sqrt{n_{calib}})$ rate, in contrast to the val-tuned protocol whose bias is $\Theta(1)$. The empirical $+0.61$ macro-F1 inflation documented in §6.2 is exactly this artifact; a half-page proposition with an appendix proof formalizes it.

Next Steps Toward an AAAI Submission

The current report establishes a clean baseline and an honest evaluation protocol, but the empirical margin reported in Section 6 is too narrow to defend, and the contribution as framed (“GNN adapter beats MLP adapter on CheXpert”) is below the bar for a generalist top-tier venue. This section specifies the contribution, architecture, training mechanism, evaluation, tables, figures, and analyses required to reach an AAAI-grade submission. The plan optimizes purely for paper quality; sequencing and runway are deliberately left out.

Reframing the Contribution

The submission reorganizes around a single thesis: *compositional concept reasoning over frozen vision–language models, with faithfulness guarantees and test-time site adaptation*. Five concrete contributions follow from this thesis.

1. A **hierarchical compositional concept architecture** (primitives $\rightarrow$ compositional layer $\rightarrow$ findings) that exploits clinical compositional structure absent in concept-bottleneck models and absent in label-graph GNNs.
2. A **faithfulness training mechanism** combining sparse causal gating with intervention-consistency loss, providing per-finding interpretability that holds up under counterfactual perturbation.
3. A **test-time adaptation recipe** that adapts only the final readout to a new clinical site, exploiting the cross-site invariance of the primitive layer.

4. A **cross-site evaluation protocol** over four chest-X-ray datasets plus one non-medical multi-label benchmark, with concept localization scored against ground-truth bounding boxes where available.
5. A **leakage-free calibration protocol** generalized as a reusable methodological contribution, with a formal consistency proposition.

The thesis survives the standard reviewer objections directly: *Why not LoRA?* cross-site performance at $< 0.1\%$ of the parameter count, statistically verified; *Why not Q-Former / CBM?* ablation rows in the headline table plus a faithfulness comparison; *Is it just calibration?* a calibration-only baseline carried throughout; *Does it generalize?* four chest-X-ray sites plus one non-medical benchmark.

Architecture: Hierarchical Compositional Concept Transformer

The adapter is a three-layer transformer-style head that consumes (i) the full sequence of frozen image tokens from a vision encoder and (ii) the frozen scalar logits / probabilities of the VLM zero-shot scorer. We refer to this head as the *Compositional Concept Adapter (CCA)* throughout.

Layer 1 — Primitive concepts. Approximately 30–50 radiological primitives (opacity, consolidation pattern, pleural effusion, volume loss, mediastinal shift, cardiac silhouette enlargement, costophrenic-angle blunting, and so on) are instantiated as learnable query embeddings, initialized from text-encoder embeddings of their canonical names and from RadGraph node embeddings where available (?). Each primitive query attends, via a single multi-head cross-attention layer, over the frozen image tokens from BiomedCLIP, RAD-DINO, or an MAE-CXR variant. Outputs are a scalar activation per primitive and a per-primitive spatial attention map.

Layer 2 — Compositional reasoning. A two-layer self-attention encoder operates on the P primitive activations augmented with a learnable compositional bias whose initialization is the RadGraph relation matrix. The bias is differentiable but constrained by the faithfulness mechanism in §.

Layer 3 — Findings readout. The C target findings (e.g., the seven or fourteen CheXpert labels) are read out from Layer 2 via attention, with the frozen VLM’s logits entering as a residual / gating signal: the adapter may override the VLM only where concept evidence is strong. Final per-class probabilities pass through the leakage-free four-way calibration protocol of §6.

Differentiation from prior art. The CCA differs structurally from Q-Former (?) (an information bottleneck without explicit composition), from CBM variants (?, ?, ?) (compositionally flat), from ML-GCN (?) (label graph only, no primitive grounding), and from KAD (?) (dual-encoder zero-shot, not a frozen-adapter head). The architecture is

generic across VLMs and modalities; chest X-ray is the primary evaluation domain, not the only one.

Faithfulness Mechanism

This subsection is the technical core of the paper. Two components.

Concept-disentangled sparse gating. A learnable binary gating matrix $M \in \{0, 1\}^{C \times P}$ specifies which primitives each finding may depend on. We relax to $\tilde{M} \in [0, 1]^{C \times P}$ via Gumbel-softmax (?) and impose an L_0 / Hoyer sparsity prior (?) with target density between 5% and 15%. At convergence each finding depends on a small, named subset of primitives, which can be read off and verified against radiologist consensus.

Causal-intervention training. During training, at each step we sample a primitive p and a counterfactual activation c' , compute downstream finding predictions $f(c_{p \leftarrow c'})$, and apply

$$\mathcal{L}_{faithful} = E_{p, c'} \|f(c_{p \leftarrow c'}) - f(c)\|_2^2 \cdot 1[\tilde{M}_{:,p} \approx 0],$$

penalizing downstream change exactly when the gating says there should be none. This is the do-calculus / Causal-CBM idea (?) adapted for compositional concepts. Total training objective: $\mathcal{L} = \mathcal{L}_{BCE} + \lambda_{sparse} \|\tilde{M}\|_0 + \lambda_{faithful} \mathcal{L}_{faithful}$, with $\lambda_{sparse}, \lambda_{faithful}$ tuned on val only.

Test-Time Site Adaptation

At deployment on a new clinical site we freeze the primitive layer and the gating matrix M (both are site-invariant by design) and adapt only the final readout via entropy-minimization on unlabeled new-site images, combined with a marginal-rate prior keeping per-finding base rates close to the source distribution unless evidence is strong. The recipe is Tent (?) for the entropy component and BACS-style marginal anchoring (?) for the prior. Reuse the official implementations; do not reimplement.

Datasets and Evaluation Protocol

Chest X-ray datasets. CheXpert (in-distribution); MIMIC-CXR-JPG (cross-site, same label schema; PhysioNet credentialed access); NIH ChestX-ray14 (cross-site, overlapping label schema); PadChest (cross-site, different language and population); VinDr-CXR (cross-site and the source of ground-truth pathology bounding boxes for localization evaluation).

Non-medical replication. At least one of MS-COCO multi-label (?) or NUS-WIDE (?), using a different frozen VLM (LLaVA, Qwen2-VL, or InternVL) to demonstrate that the method is not chest-X-ray-specific.

Evaluation protocol. The four-way split (`train / val / calib / test`) of §6 is retained. All hyperparameters and calibration thresholds are tuned on `val` and `calib` respectively; `test` is touched exactly once per model. Cross-site evaluation uses the source-site `calib` for threshold initialization, then optionally re-tuned on a held-out fraction of the target site only for TTA experiments.

Metrics

- **Discrimination.** Per-class and macro AUROC and AUPRC (`torchmetrics.classification.MultilabelAUROC` cross-checked against `sklearn.metrics.roc_auc_score`).
- **Operating-point performance.** F1, precision, recall at the calibrated threshold per class.
- **Calibration.** ECE and Brier score per class, plus reliability diagrams in the appendix.
- **Faithfulness.** Held-out intervention-consistency score (the $\mathcal{L}_{faithful}$ surrogate on a held-out intervention set) and a complementary necessity / sufficiency score (?).
- **Concept localization.** Pointing-game accuracy and IoU between primitive attention maps and ground-truth bounding boxes on VinDr-CXR and the annotated subset of CheXpert.
- **TTA gain.** Δ AUROC pre- vs post-adaptation, per cross-site dataset pair.
- **Cost.** Total trainable parameters; GPU-hours to train; inference latency (median, p95) on a fixed reference batch; peak GPU memory.

Statistical reporting. Every number is reported as mean $\pm$ standard deviation across five seeds, with a paired bootstrap 95% CI on the test predictions (1,000 resamples). AUROC differences between the proposed method and each baseline are tested with DeLong’s test. Across-label multiple testing is corrected with Benjamini–Hochberg at $q = 0.05$.

Pre-registered success criterion. The submission’s headline claim is: *CCA outperforms LoRA-16 on cross-site macro-AUROC by a margin whose paired bootstrap 95% CI excludes zero, at $< 0.1\%$ of the trainable parameter count, on at least three of the four cross-site dataset pairs.* This criterion is fixed before the cross-site experiments are run.

Baseline Matrix

Twenty-five rows. Anything fewer invites a missing-baseline rejection.

Frozen-output adapters. Zero-shot VLM; linear probe on $[z; p]$; MLP on $[z; p]$; MLP on $[z; p; e_{CLIP}]$; ML-GCN (?); Q-Former-style learnable-query adapter (?); post-hoc CBM (?); Label-free CBM (?); the current bipartite head (as the prior version of this work).

PEFT on the VLM. LoRA (?) at ranks $\{4, 8, 16, 32\}$; LoRA on the vision tower only; prefix tuning (?); visual prompt tuning (?); BitFit (?).

Medical-domain baselines. BiomedCLIP linear probe (?); RAD-DINO linear probe (?); KAD re-run on our splits with the official code (?); CXR-LLaVA (?) and MAIRA-2 (?) where checkpoints are public.

Loss and calibration baselines. Asymmetric loss (?); logit adjustment (?); temperature scaling; per-class isotonic regression (`sklearn.isotonic`); deep ensembles (?).

Test-time adaptation baselines. Tent (?); MEMO (?); source-free DA (?).

Theory

Two propositions, both with full proofs in the appendix and informal statements in the main text. Half a page each.

Proposition 1 (Faithfulness–utility tradeoff). Under the sparse-gating prior with L_0 budget k , the achievable region of (predictive AUROC, intervention-faithfulness score) lies on a Pareto frontier whose curvature is bounded by the spectral gap of the primitive-finding cross-covariance matrix. The proposition characterizes when sparsifying the gating costs accuracy and when it does not.

Proposition 2 (Sufficient-statistic characterization). State and prove conditions under which the frozen VLM scores (z, p) form a sufficient statistic for the multi-label posterior $\Pr(y \mid x)$, versus when the raw image tokens carry residual predictive signal. The compositional structure of y shifts this boundary; the proposition connects directly to the empirical observation in §6 that calibrated zero-shot and the MLP adapter are indistinguishable, while a structured concept head with image-token access is not.

Proposition 3 (Calibration consistency, formalizing §6.2). The four-way leakage-free protocol yields unbiased per-class threshold estimates whose error decays at the expected $O(1/\sqrt{n_{calib}})$ rate, while the val-tuned protocol has bias $\Theta(1)$. The empirical $+0.61$ macro-F1 inflation in §6.2 is exactly this artifact.

Interpretability Deliverables

Three pieces; all sit in the main paper, not the appendix.

Radiologist evaluation. Two to three radiologists score 100 randomly sampled CheXpert test cases on a 5-point Likert scale of clinical plausibility of the primitive activations and attention maps. The CCA is compared against (i) a Grad-CAM baseline on a fine-tuned ResNet-50 and (ii) the Q-Former-style adapter. Inter-rater agreement reported via Krippendorff’s α .

Intervention case studies. Ten clinically interesting cases. For each, intervene on one primitive (e.g., force “opacity” off) and show that downstream finding predictions change in the directions a radiologist would predict. Visualize as a causal trace.

Failure-mode taxonomy. A held-out collection of 50–100 failure cases, manually categorized into recurring failure modes (e.g., occluded findings, atypical presentations, dataset-label noise), with primitive attention maps shown.

Tables

This subsection enumerates every table the submission must include. Specifications are precise: row labels, column labels, statistical decorations, and footnote requirements.

Main paper

- **Table 1 — Headline cross-site results.** Rows: CCA (ours), the five strongest baselines (zero-shot, LoRA-16, Q-Former, CBM, KAD). Columns: in-distribution macro-AUROC on CheXpert; cross-site macro-AUROC on each of MIMIC-CXR, NIH, PadChest, VinDr-CXR; trainable parameter count; inference latency (ms). Every cell is mean $\pm$ std over five seeds; CCA vs each baseline difference annotated with paired-bootstrap-CI inclusion of zero and DeLong p -value. Best per column in bold; statistically tied for best underlined. Footnote specifies multiple-testing correction.
- **Table 2 — In-distribution CheXpert results, full baseline matrix.** All twenty-five baseline rows. Columns: macro-AUROC, macro-AUPRC, macro-F1 (calibrated), ECE, Brier, parameter count, GPU-hours.
- **Table 3 — Test-time adaptation gains.** Rows: CCA, three TTA baselines (Tent, MEMO, SHOT) applied to LoRA-16 and to the MLP adapter. Columns: Δ macro-AUROC for each cross-site direction, paired with statistical significance markers.
- **Table 4 — Faithfulness metrics.** Rows: CCA at three sparsity levels, post-hoc CBM, Label-free CBM, Q-Former. Columns: intervention-consistency score, necessity, sufficiency, primitive localization pointing-game accuracy, IoU on VinDr-CXR.
- **Table 5 — Radiologist evaluation.** Rows: CCA, Grad-CAM-on-ResNet-50, Q-Former. Columns: mean Likert plausibility, Krippendorff’s α , percentage of cases rated ≥ 4 . Per-rater scores in an appendix table.
- **Table 6 — Non-medical replication on MS-COCO multi-label.** Rows: CCA, LoRA-16, Q-Former, ML-GCN, MLP. Columns: macro-mAP, macro-F1, parameter count. Demonstrates that the method is not chest-X-ray-specific.
- **Table 7 — Cost summary.** Rows: each method in Table 1. Columns: trainable params; total params; GPU-hours to train; inference latency p50/p95; peak GPU memory; FLOPs per forward pass. This is the table that anchors the deployment-cost claim.

Appendix

- **Table A.1** — Per-class AUROC and AUPRC on CheXpert test, for every baseline and CCA, with 95% CIs.
- **Table A.2** — Per-class results on MIMIC-CXR, NIH, PadChest, VinDr-CXR (one table per dataset, identical structure).
- **Table A.3** — Full ablation matrix: removing primitive layer, removing compositional layer, removing residual VLM gating, removing sparsity prior, removing intervention loss, removing TTA, swapping image encoder (CLIP-B/32, BiomedCLIP, RAD-DINO, MAE-CXR), varying primitive count $P \in \{15, 30, 50, 80\}$, varying sparsity target.
- **Table A.4** — Permuted-graph control: CCA with the Rad-Graph prior randomly shuffled. If results are unchanged, the prior is doing nothing.
- **Table A.5** — Held-out-concept generalization: mask 20% of findings at train time, evaluate zero-shot AUROC on the held-out findings.
- **Table A.6** — Per-class ECE and Brier on every dataset.
- **Table A.7** — DeLong test pairwise p -value matrix between CCA and every baseline, per dataset.
- **Table A.8** — Per-seed raw numbers for the headline results (Table 1), so reviewers can verify variance.
- **Table A.9** — Hyperparameter table for every method (learning rate, batch size, epochs, weight decay, scheduler, loss weights, λ ’s, sparsity target).
- **Table A.10** — Compute cost breakdown by phase (feature extraction, adapter training, TTA inference).
- **Table A.11** — Per-rater radiologist scores with confidence intervals; pairwise rater agreement matrix.
- **Table A.12** — Failure-mode taxonomy: count and percentage of each failure category, per dataset.
- **Table A.13** — Calibration policy comparison (U-Zeros, U-Ones, U-Ignore) on CheXpert.
- **Table A.14** — Effect of the four-way protocol vs the val-tuned protocol (Proposition 3, empirical version).

Figures

Main paper

- **Figure 1 — Pull figure.** Left: schematic of the deployment scenario (hospital with frozen API access + labeled data). Right: cross-site macro-AUROC vs trainable parameter count for CCA and the four strongest baselines, with CCA on the Pareto frontier. Single figure, conveys the entire paper.
- **Figure 2 — Architecture diagram.** The three layers (primitives $\rightarrow$ compositional $\rightarrow$ findings), with frozen blocks shaded grey, learnable blocks coloured, and the residual VLM gating shown as a separate path.
- **Figure 3 — Qualitative concept-attention maps.** Six representative cases (one normal, five with different pathologies), each with the relevant primitive’s spatial

attention overlaid on the X-ray. Ground-truth bounding boxes (where available) shown in dashed outline.

- **Figure 4 — Faithfulness–utility Pareto frontier.** X-axis: intervention-consistency score; Y-axis: macro-AUROC. CCA at varying sparsity levels traces a curve; baseline methods are points off the frontier.
- **Figure 5 — Intervention case study.** A single case with primitive activations on the left, the finding readout on the right, and arrows showing the downstream change when one primitive is forced to a counterfactual value.
- **Figure 6 — Calibration reliability diagrams.** One sub-panel per finding for CheXpert test, comparing CCA pre- and post-calibration against LoRA-16.
- **Figure 7 — Test-time adaptation dynamics.** AUROC on the target site as a function of TTA optimization steps, for CCA vs Tent-on-LoRA vs MEMO-on-LoRA. Shaded bands are ± 1 std across seeds.

Appendix

- **Figure A.1 —** The learned gating matrix M before and after training, as a heatmap. Demonstrates emergence of named compositional structure.
- **Figure A.2 —** Sparsity convergence curves over training epochs.
- **Figure A.3 —** Per-class confusion matrices on CheXpert, MIMIC, NIH, PadChest.
- **Figure A.4 —** Extended qualitative gallery: 40 cases with primitive attention overlays.
- **Figure A.5 —** Failure-mode gallery: 20 failure cases with attention maps and failure-mode labels.
- **Figure A.6 —** t-SNE / UMAP of the learned primitive embeddings, coloured by clinical category, demonstrating semantic clustering.
- **Figure A.7 —** Hyperparameter sensitivity heatmaps for λ_{sparse} , $\lambda_{faithful}$, learning rate.
- **Figure A.8 —** Training loss curves for the headline runs.
- **Figure A.9 —** Radiologist Likert score distributions, per method and per rater.
- **Figure A.10 —** Cumulative AUROC vs training-set size (data-efficiency curves) for CCA and the strongest fine-tuning baseline.

Analyses

Each analysis is at least one paragraph of main-paper prose, supported by one of the tables or figures above.

Headline cross-site analysis. Quantify the gain over LoRA-16 per cross-site direction; identify which findings drive the gain; correlate gain with primitive activation strength on the target site.

Ablation analysis. Walk through each ablation row of Table A.3; identify the single component whose removal costs the most (expected: image-token access via the primitive layer); identify any components that can be removed without loss (expected: a candidate for paper simplification).

Faithfulness analysis. Show that CCA’s intervention-consistency is significantly higher than CBM variants; show that this comes at ≤ 0.5 AUROC cost on CheXpert; argue that the tradeoff falls on the favourable side of the Pareto frontier characterized by Proposition 1.

Concept-emergence analysis. Examine the gating matrix M at convergence; for each finding, list the top- k primitives it gates; assess clinical plausibility with the radiologist panel; report agreement rate.

TTA analysis. Decompose the cross-site gain into (i) source-only zero-shot, (ii) source-only with calibration transfer, (iii) full TTA. Identify which sites benefit most and hypothesise why (label-prior shift vs covariate shift).

Localization analysis. On VinDr-CXR and CheXpert with annotations, compute pointing-game accuracy and IoU per primitive; compare against Grad-CAM and Q-Former attention maps.

Cost analysis. Argue from Table 7 that the deployment cost story holds: $< 0.1\%$ of LoRA-16 trainable parameters, $\leq 5\%$ of LoRA-16 inference FLOPs, no VLM gradient access required.

Failure analysis. Categorize failures using the taxonomy in Table A.12; argue that the majority of failures are dataset-label-noise or atypical-presentation cases rather than method-specific failure modes.

Statistical-rigor section. Explicitly state the statistical protocol: five seeds, paired bootstrap CI, DeLong for AUROC, Benjamini–Hochberg multiple-testing correction, pre-registered success criterion. This section pre-empts a non-trivial fraction of reviewer concerns.

Engineering Discipline

The implementation collapses the per-script duplication noted in §8.3 into a single `common_train.py` (training loop, metrics, registry interaction) and a `models/` package with one file per architecture. Run-tracking is unchanged from the current registry. Calibration runs as a post-hoc stage with its own dedicated module. Features are extracted once and persisted as `.npz` / `.pt`, not JSON, with on-disk caches keyed by (dataset, encoder, preprocessing-version). Multi-seed runs are launched through a thin scheduler that pins seeds, records git hashes, and emits one row per run into a results parquet consumed by the table-generation scripts. Every result in the paper is reproducible from a single `make reproduce`.

Risks and Contingencies

If the headline cross-site gain over LoRA-16 fails to clear the pre-registered margin, the paper retargets around faithfulness: the title and Section 1 reframe as “a faithful compositional adapter that matches PEFT cross-site at fraction-of-the-parameters cost,” and the cost / faithfulness story carries the contribution. If the faithfulness mechanism fails to converge cleanly (the sparsity prior collapses or the intervention loss destabilizes), the recovery path is the architecture without intervention loss, with faithfulness scored post-hoc; this is weaker but still publishable. If radiologist evaluation is infeasible, the localization-against-bounding-boxes metric on VinDr-CXR substitutes as a quantitative proxy for clinical plausibility, with the radiologist study deferred to a journal extension.

References

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Journal Article

Robinson, A. L. 1980a. New Ways to Make Microcircuits Smaller. *Science* 208: 1019–1026.

Magazine Article

Hasling, D. W.; Clancey, W. J.; and Rennels, G. R. 1983. Strategic Explanations in Consultation. *The International Journal of Man-Machine Studies* 20(1): 3–19.

Proceedings Paper Published by a Society

Clancey, W. J. 1983b. Communication, Simulation, and Intelligent Agents: Implications of Personal Intelligent Machines for Medical Education. In *Proceedings of the Eighth International Joint Conference on Artificial Intelligence*, 556–560. Menlo Park, Calif.: International Joint Conferences on Artificial Intelligence, Inc.

Proceedings Paper Published by a Press or Publisher

Clancey, W. J. 1984. Classification Problem Solving. In *Proceedings of the Fourth National Conference on Artificial Intelligence*, 49–54. Menlo Park, Calif.: AAAI Press.

University Technical Report

Rice, J. 1986. Poligon: A System for Parallel Problem Solving, Technical Report, KSL-86-19, Dept. of Computer Science, Stanford Univ.

Dissertation or Thesis

Clancey, W. J. 1979b. Transfer of Rule-Based Expertise through a Tutorial Dialogue. Ph.D. diss., Dept. of Computer Science, Stanford Univ., Stanford, Calif.

Forthcoming Publication

Clancey, W. J. 1986a. The Engineering of Qualitative Models. Forthcoming.

Producing Reliable PDF Documents with L^AT_EX

Generally speaking, PDF files are platform independent and accessible to everyone. When creating a paper for a proceedings or publication in which many PDF documents must be merged and then printed on high-resolution PostScript RIPs, several requirements must be met that are not normally of concern. Thus to ensure that your paper will look like it does when printed on your own machine, you must take several precautions:

- Use type 1 fonts (not type 3 fonts)
- Use only standard Times, Nimbus, and CMR font packages (not fonts like F3 or fonts with tildes in the names or fonts—other than Computer Modern—that are created for specific point sizes, like Times~19) or fonts with strange combinations of numbers and letters
- Embed all fonts when producing the PDF
- Do not use the [T1]fontenc package (install the CM super fonts package instead)

Creating Output Using PDFL^AT_EX Is Required

By using the PDFL^AT_EX program instead of straight L^AT_EX or T_EX, you will probably avoid the type 3 font problem altogether (unless you use a package that calls for metafont). PDFL^AT_EX enables you to create a PDF document directly from L^AT_EX source. The one requirement of this software is that all your graphics and images must be available in a format that PDFL^AT_EX understands (normally PDF).

PDFL^AT_EX's default is to create documents with type 1 fonts. If you find that it is not doing so in your case, it is likely that one or more fonts are missing from your system or are not in a path that is known to PDFL^AT_EX.

dvipdf Script Scripts such as dvipdf which ostensibly bypass the Postscript intermediary should not be used since they generally do not instruct dvips to use the config.pdf file.

dvipdfm Do not use this dvi-PDF conversion package if your document contains graphics (and we recommend you avoid it even if your document does not contain graphics).

Ghostscript

L^AT_EX users should not use GhostScript to create their PDFs.

Graphics

If you are still finding type 3 fonts in your PDF file, look at your graphics! L^AT_EX users should check all their imported graphics files as well for font problems.

Proofreading Your PDF

Please check all the pages of your PDF file. Is the page size A4? Are there any type 3, Identity-H, or CID fonts? Are all the fonts embedded? Are there any areas where equations or figures run into the margins? Did you include all your figures? Did you follow mixed case capitalization rules for your title? Did you include a copyright notice? Do any of the pages scroll slowly (because the graphics draw slowly on the page)? Are URLs underlined and in color? You will need to fix these common errors before submitting your file.

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L^AT_EX 209 Warning

If you use L^AT_EX 209 we will not be able to publish your paper. Convert your paper to L^AT_EX 2e.

Naming Your Electronic File

We request that you name your L^AT_EX source file with your last name (family name) so that it can easily be differentiated from other submissions. If you name your files with the name of the event or “aaai” or “paper” or “camera-ready” or some other generic or indecipherable name, you bear all risks of loss — it is extremely likely that your file may be overwritten.

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If you have any questions about the preparation or submission of your paper as instructed in this document, please contact AAAI Press at the address given below. If you have technical questions about implementation of the aaai style file, please contact an expert at your site. We do not provide technical support for L^AT_EX or any other software package. To avoid problems, please keep your paper simple, and do not incorporate complicated macros and style files.

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Additional Resources

L^AT_EX is a difficult program to master. If you’ve used that software, and this document didn’t help or some items were not explained clearly, we recommend you read Michael Shell’s excellent document (testflow doc.txt V1.0a 2002/08/13) about obtaining correct PS/PDF output on L^AT_EX systems. (It was written for another purpose, but it has general application as well). It is available at www.ctan.org in the tex-archive.

Acknowledgments

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Thank you for reading these instructions carefully. We look forward to receiving your electronic files!